Research Methodology

Module 6

Research Reporting and Ethics

Prof. Subendu Saha

Compiled by Amitendu and Claude

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1 Structure of a Research Report, Thesis, or Paper

1.1 Purpose of Reporting

A research report **communicates the study** clearly so that others can understand, evaluate, and repeat it. Research that is not reported is of no use to the scientific community.

1.2 Types of Research Writing

Type	Description	Typical length
Research report	Detailed account of a study, often for a sponsor or institution	Medium to long
Thesis / dissertation	Full-length work for a degree (M.Phil., Ph.D., Master's)	Long, many chapters
Research paper / article	Concise account for a journal or conference	Short (about 4,000–8,000 words)

1.3 Structure of a Thesis or Report

Part	Contents
Preliminary pages	Title page, certificate, declaration, acknowledgements, abstract, table of contents, lists of tables and figures
Chapter 1	Introduction: background, problem, objectives, scope
Chapter 2	Review of literature and research gap
Chapter 3	Research methodology: design, sample, tools, data collection, analysis plan
Chapter 4	Results and analysis
Chapter 5	Discussion, conclusions, recommendations, limitations
End matter	References/bibliography, appendices (questionnaire, raw tables)

1.4 Structure of a Research Paper (IMRaD)

Most journal papers follow the **IMRaD** format:

1. **I**ntroduction: what is the problem and why does it matter?
2. **M**ethods: how was the study done?
3. **R**esults: what was found?
4. **a**nd **D**iscussion: what do the findings mean?

A paper also has a **title, abstract, keywords**, conclusion, **acknowledgements**, and **references**.

> **Exam Tip** Write the **abstract last**, even though it appears first. It is easiest to summarise once everything else is complete.

1.5 Characteristics of a Good Report

- **Clear and concise** language; avoid jargon and padding.
- **Logical order** with headings and sub-headings.
- **Objective** tone, usually in the third person and past tense.
- **Accurate** data, tables, and figures, each numbered and referred to in the text.
- **Consistent** format, spelling, and citation style.
- Free from **grammar and typing errors**.

2 Writing the Main Sections

2.1 Abstract

A **short, self-contained summary** of the whole study, usually 150–250 words. Many readers decide from the abstract alone whether to read further.

- **Include:** the problem and aim, method, main results, and main conclusion.
- **Exclude:** citations, tables, and long background.
- Add 4–6 **keywords** for indexing and search.

2.2 Introduction

Sets the stage and **moves from general to specific** (like a funnel).

1. Background and context of the topic.
2. Brief review of existing work and the **research gap**.
3. **Statement of the problem.**
4. **Objectives, research questions, or hypotheses.**
5. Significance and scope; outline of the rest of the paper (optional).

2.3 Methodology

Explains **exactly how** the study was done so that another researcher could repeat it.

- Research design and approach (experimental, descriptive, etc.).
- Population, sampling method, and sample size.
- Tools used (questionnaire, interview guide, instruments) and their reliability and validity.
- Procedure for data collection.
- Methods of data analysis and software used.
- Ethical considerations (consent, confidentiality, approval).

> **Note** Write the methodology in the **past tense** and give enough detail to make the study **replicable**.

2.4 Results

Presents the **findings only**, without personal opinion.

- Follow the order of the research questions or objectives.
- Use **tables and figures** for clarity, and describe the key points in the text.
- Report test statistics, p-values, and effect sizes properly.
- Do not repeat every number from the tables in the text.

> **Watch Out** Do not mix **results** with **discussion**. Results say *what was found*; discussion says *what it means*.

2.5 Discussion

Interprets the results, compares them with earlier studies, explains unexpected findings, and notes limitations.

2.6 Conclusion

A **brief closing** that answers the research questions and states the overall message.

- Summarise the main findings and link them to the objectives.
- State the contribution of the study and practical implications.
- Give recommendations, limitations, and directions for future research.
- Do **not** introduce new data or new references.

Section	Key question it answers
Abstract	What was done and found, in brief?
Introduction	Why was the study needed?
Methodology	How was it done?
Results	What was found?
Discussion	What does it mean?
Conclusion	What is the final message?

3 Referencing Styles (APA, MLA, IEEE)

3.1 Why Reference?

- Gives **credit** to the original authors and avoids plagiarism.
- Lets readers **trace and verify** sources.
- Shows the **depth and credibility** of your work.

A **citation** appears inside the text; a **reference list** at the end gives full details. A **bibliography** may also list works read but not cited.

3.2 APA (American Psychological Association, 7th ed.)

Common in psychology, education, and social sciences. Uses the **author–date** system.

- **In-text:** (Sharma & Das, 2022) or Sharma and Das (2022).
- **Order:** reference list in **alphabetical** order by author surname.

Journal article:
Sharma, R., & Das, A. (2022). Smartphone use and academic performance. *Journal of Education Studies, 12*(3), 45–58. https://doi.org/xxxx

Book:
Kothari, C. R. (2004). *Research methodology: Methods and techniques* (2nd ed.). New Age International.

3.3 MLA (Modern Language Association, 9th ed.)

Common in literature, languages, and the humanities. Uses the **author–page** system.

- **In-text:** (Sharma and Das 45), with no comma and no "p."
- **List:** titled "Works Cited," in alphabetical order.

Journal article:
Sharma, Rina, and Arun Das. "Smartphone Use and Academic Performance." *Journal of Education Studies*, vol. 12, no. 3, 2022, pp. 45–58.

Book:
Kothari, C. R. *Research Methodology: Methods and Techniques*. 2nd ed., New Age International, 2004.

3.4 IEEE (Institute of Electrical and Electronics Engineers)

Common in engineering, computer science, and technology. Uses **numbered** citations.

- **In-text:** a number in square brackets, such as [1].
- **List:** ordered by **first appearance** in the text, not alphabetically.

Journal article:
[1] R. Sharma and A. Das, "Smartphone use and academic performance," *J. Educ. Stud.*, vol. 12, no. 3, pp. 45–58, 2022.

Book:
[2] C. R. Kothari, *Research Methodology: Methods and Techniques*, 2nd ed. New Delhi, India: New Age International, 2004.

3.5 Quick Comparison

	APA	MLA	IEEE
Field	Social sciences	Humanities	Engineering, IT
In-text	(Author, Year)	(Author Page)	[Number]
List name	References	Works Cited	References
List order	Alphabetical	Alphabetical	By appearance
Year position	After author	Near the end	At the end

> **Exam Tip** Pick **one style** and use it consistently. Follow your university or journal guidelines. Reference managers such as **Zotero**, **Mendeley**, and **EndNote** format citations automatically.

4 Plagiarism and Research Ethics

4.1 Plagiarism

> **Plagiarism** Presenting **someone else's words, ideas, data, or work as your own** without proper acknowledgement.

Forms of Plagiarism

- **Direct (copy–paste):** copying text word for word without quotation marks or citation.
- **Paraphrasing plagiarism:** changing a few words but keeping the original structure and idea without credit.
- **Mosaic (patchwork):** stitching together phrases from several sources.
- **Self-plagiarism:** reusing your own earlier published work without disclosure.
- **Idea plagiarism:** using another person's original idea or argument without credit.
- **Accidental:** careless or incorrect citation or poor note-taking.
- **Data plagiarism or purchased work:** using others' data, or having someone else write the work.

How to Avoid Plagiarism

- **Cite** every idea, fact, figure, or quotation that is not your own.
- Use **quotation marks** for exact words, plus the citation.
- **Paraphrase** properly: understand it, rewrite in your own words and structure, and still cite.
- Keep careful **notes with source details** while reading.
- Use a **plagiarism checker** such as Turnitin or iThenticate before submission.
- Know your institution's **academic integrity** rules and permitted similarity limits.

> **Note** A similarity score is only a **screening tool**. A low score does not prove originality, and a high score may come from properly cited quotations and common phrases. Judgement is still needed.

4.2 Research Ethics

> **Research Ethics** The **moral principles and standards of conduct** that guide how research is planned, carried out, and reported, so as to protect participants and preserve the integrity of knowledge.

Principle	Meaning
Informed consent	Participants are told the purpose, procedure, and risks, and agree freely.
Voluntary participation	No pressure; participants may withdraw at any time.
Confidentiality	Personal information is protected and shared with no one without permission.
Anonymity	Identities are not recorded or revealed.
No harm	Avoid physical, mental, social, or financial harm to participants.
Honesty and integrity	Report methods and results truthfully.
Objectivity	Avoid bias; disclose conflicts of interest.
Fair authorship	Credit all who contributed, and only them.
Respect for vulnerable groups	Extra care with children, patients, and other vulnerable people.
Data protection	Store and handle data securely.

Research Misconduct (FFP)

- **Fabrication:** making up data or results.
- **Falsification:** manipulating data, equipment, or processes to change results.
- **Plagiarism:** using others' work without credit.

Other violations include selective reporting, duplicate publication, misleading authorship, and ignoring conflicts of interest.

Ethical Review

Studies involving humans or animals are usually reviewed by an **Institutional Ethics Committee (or Review Board)** before data collection. Ethical clearance is often required for thesis work and journal publication.

> **Watch Out** Misconduct can lead to **rejection or retraction** of papers, loss of degree, damaged reputation, and legal action.

5 Preparing a Research Proposal and Presentation

5.1 Research Proposal

A research proposal is a **plan submitted before the study** to show what will be done, why it matters, and how it will be done. It is used to get approval, admission, or funding.

Components of a Proposal

1. **Title:** short, specific, and informative.
2. **Introduction and background:** context and importance.
3. **Statement of the problem** and research gap.
4. **Objectives, research questions, hypotheses.**
5. **Review of literature:** brief and focused.
6. **Research methodology:** design, sample, tools, data collection, analysis, ethical issues.
7. **Expected outcomes** and significance of the study.
8. **Work plan or timeline** (often a Gantt chart).
9. **Budget** (if funding is sought) and resources needed.
10. **References.**

> **Exam Tip** A strong proposal convinces the reader of three things: the problem is **important**, the plan is **feasible**, and the researcher is **capable**.

Common Weaknesses

- Vague title, problem, or objectives.
- Topic too broad or unrealistic for the time and resources.
- Poor link between objectives and methods.
- No research gap, or outdated literature.
- Missing ethics and timeline.

5.2 Research Presentation

Presenting research orally (viva, seminar, conference) needs **clarity and selection**: show the essence, not everything.

Typical Slide Structure

1. Title and presenter details.
2. Background and problem.
3. Objectives and research questions.
4. Methodology.
5. Key results (charts and tables).
6. Conclusion and recommendations.
7. Limitations and future work.
8. Acknowledgements and questions.

Tips for Effective Presentation

- **One main idea per slide**; use short points, not paragraphs.
- Use **large, readable fonts** and good contrast; avoid clutter.
- Prefer **charts and visuals** to dense tables.
- Keep to the **time limit**; as a rough guide, about one slide per minute.

- **Practise** aloud, and prepare for likely questions.
- Maintain eye contact and speak clearly instead of reading the slides.
- Be honest about limitations when answering questions.

6 Quick Revision Sheet

- **Thesis:** preliminary pages, chapters (introduction, literature, methodology, results, conclusion), references, appendices.
- **IMRaD:** Introduction, Methods, Results, and Discussion.
- **Abstract:** 150–250 words; problem, method, results, conclusion; write it last.
- **Results** state what was found; **discussion** explains what it means.
- **APA:** (Author, Year), alphabetical. **MLA:** (Author Page), Works Cited. **IEEE:** [1], by order of appearance.
- **Plagiarism:** using others' work without credit; avoid by citing, quoting, and paraphrasing properly.
- **Ethics:** consent, confidentiality, no harm, honesty, fair authorship.
- **Misconduct (FFP):** fabrication, falsification, plagiarism.
- **Proposal:** title, problem, objectives, literature, methodology, timeline, budget, references.
- **Presentation:** simple slides, visuals, practised delivery, within the time.

End of Module 6